

Stress and Thermal Analysis of CubeSat Structure

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Keywords: CubeSat, ANSYS, stress, thermal

Abstract. This paper presents the stress and thermal analysis on the CubeSat structure to study the survivability of the CubeSat during the launching process or operating condition at the orbit is presented. Various design of mechanical structures were analyzed to determine the best design for different mission requirements. Analysis on the temperature of the batteries will be conducted as it is one of the most critical components that must operate in the required temperature to avoid failure of the CubeSat. ANSYS 13.0 was used to simulate both the structural and thermal analysis. Static structural was used to study the impact of G-force on the CubeSat during the launching process and Icepak was used to study the internal temperature. All of the result will be compiled in the table and comparisons were made among different designs to determine the advantages and disadvantages of each design. Results from simulation such as: safety factor, weight, internal available space and battery discharge rate were analyzed. From the findings, there is no best design in the CubeSat structure but only the most suitable design for the mission purposes. Battery discharge rate will play an important role to determine the requirement of heater in CubeSat.

Introduction

CubeSat is cube shape satellite with the dimension of 10 x 10 x 10 cm. The small size and shape of CubeSat promises more flexibility and success as the tasks were distributed to each CubeSat. With the distribution of task, the failure of a single CubeSat will not cause a critical failure to the whole system. Whenever a CubeSat fails, the rest can still operatenormally until replacement is launched. The basic CubeSat design specifications ensure the safety of the launch vehicle, CubeSat, and any other device in the rocket. The objective is to ensure that if a CubeSat were to fail as a result of a poor or incompatible design choice, it would not cause a failure of the entire launch system [1]. Various types of rocket can be used for launching the CubeSat to the orbit and it is important for CubeSat developer to determine which rocket to be used as the CubeSat structure must be able to sustain the various condition of the rocket during the launching condition. Rocket user guide is a very important paper that must be studied and fully understood by the CubeSat developer [2,3,4]. Besides that, verification standard should also be studied by the user as it covers the launching environment for flight projects [5]. There are two types of thermal control that can be applied on a CubeSat, which is active and passive thermal control. In space, heat will be transfer from one entity to another entity in three different methods [6, 7, 8] : Radiation, conduction and convection

The three main heat sources [9, 10, 11] :

- The Earth albedo approximated by a mean value of $a = 0.33$
- The Earth IR radiation can be approximated by a 255K black body ($\zeta T_{Earth4} \sim 240 \text{ W/m}^2$)
- Direct solar flux

Methodology

Mechanical subsystem concerns the durability and strength of the CubeSat during launching to the outer space. During the process, there are mainly three types of mechanical loading acting on the CubeSat, which are: quasistatic, static and dynamic loading. For this report, it will only cover the scope of static loading acting on the CubeSat. The analysis of the static loading acting on the CubeSat will emphasize on the stress distribution and deformation of the CubeSat chassis. Due to the fact that the size and shape of CubeSat needs to fulfill certain requirements, the limitation of the design of the chassis itself is very constrained. Modification that can be done on the CubeSat chassis design is mostly on the shape of the support surface for the structure, usage of beam and etc. For initial case of the design process, the CubeSat chassis that were being modeled is 1U. In the beginning, since that the critical stress distribution acting on the CubeSat is unknown, a most basic CubeSat was modeled where the CubeSat surface consists of six thin plate as shown in Figure 1 and Figure 2.

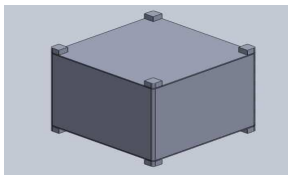


Figure 1- Initial 1U CubeSat without beam

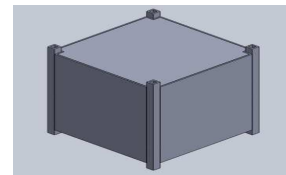


Figure 2- Initial 1U CubeSat with beam

As shown in Figure 3, the red color arrow represents the applied load onto the structure. The 30G load acting in vertical direction is divided by four giving a value of 73.575 N. This means that each of the extruded pin will be loaded by a force of 73.575 N. The alphabets of E, F, G and H at the lowermost extruded pins represent the fixed support which will be in contact with the lower boundary of P-Pod. For beam CubeSat, the load will be applied on the uppermost part of the beam and the fixed support will be set at the lowermost part of the beam. Figure 4 shows the loading condition of the beam CubeSat.

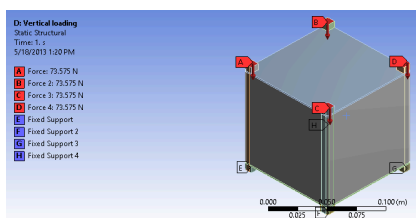


Figure 3- Loading condition of beamless

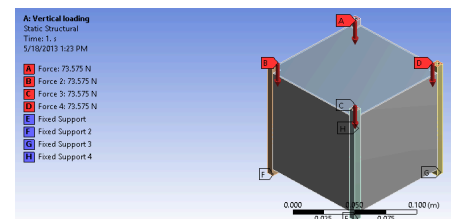


Figure 4- Loading condition of beam

As shown in Figure 5, the red color arrow represents the applied load onto the structure. The 10G load acting in horizontal direction is divided by two giving a value of 49.05 N. This means that each of the thin film will be loaded by a force of 49.05 N. The alphabet C and D at the left side thin film represent the fixed support which will be in contact with the lower boundary of P-Pod. For beam CubeSat, the load will be applied on the beam at the right and the fixed support will be set on the beam at the left. Figure 6 shows the loading condition of the beam CubeSat.

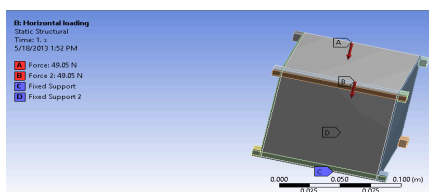


Figure 5- Loading condition of beamless

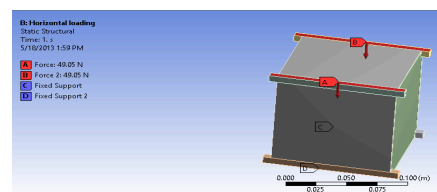


Figure 6- Loading condition of beam

Results and Discussion

(a) Vertical Load:

The highest stress acting on the structure is 9.6544 MPa and it is located at the joint between the extruded pin and the upper plate. The reason for this situation is that the thickness of the thin plate is very small, which is: 1.27 mm. The highest deformation of the structure is located on top of the pin with the value of 3.5175×10^{-6} m. This is due to the vertical load is acted directly on the extruded pin. Hence the load was distributed to other supporting structure, such as: vertical and horizontal plate when the pin is in contact with those structures. From this analysis, it shows that the extruded pin was the first component to be displaced when a load acted onto the structure.

(b) Horizontal Load:

The highest stress acting on the structure is 4.0837 MPa at the location nearing the edge of the structure as that is the location where load is being applied onto the structure. The highest deformation of the structure is located at the middle of the contact plate with the value of 1.3742×10^{-5} m. The reason for this situation to occur is due to the horizontal load is acted on the thin film which will be in contact with that surface. The load was distributed evenly on that surface before being distributed to other supporting structure which are the four plates surrounding it. For this reason, the plate in contact with thin film will be displaced to the maximum value especially at the middle of the plate as there are no supports for it at the side.

As shown in Table 1, the overall deformation of the structure increased as more holes being design on the structure. When more holes were put into the structure, it will cause the structure to be easier to deform. However the deformation value is very small as the highest is only 34.755×10^{-6} m which is very small and could be ignored.

Table 1- Compilation result for mechanical subsystem

CubeSat		Vertical Loading		
		Equivalent Stress (MPa)	Safety Factor	Deformation (e-6m)
Beamless	C1	9.6544	28.5880	3.5175
	C2	8.9616	30.7981	3.5392
	C3	8.9989	30.6704	7.3709
	C4	8.7598	31.5076	5.0463
Beam	C5	5.4149	50.9704	2.3613
	C6	5.3821	51.2811	2.3836
	C7	5.3635	51.4589	2.5495
	C8	5.2985	52.0902	2.535
		Horizontal Loading		
Beamless	C1	4.0837	67.5858	13.742
	C2	4.2064	65.6143	16.223
	C3	15.252	18.0960	34.755
	C4	10.628	25.9691	25.289

Beam	C5	1.3808	199.8841	1.654
	C6	1.339	206.1240	1.8322
	C7	2.6921	102.5222	2.6891
	C8	4.079	67.6636	2.3681

Table 2 shows the advantage and disadvantage of each CubeSat design. Each of the CubeSat design will have its own advantage and there is no best design in the CubeSat structure but only the most suitable design for the mission purposes. When designing the CubeSat structure, the designer should choose the most suitable design for the CubeSat mission referring to the result from Table 3.

Table 2- Ranking of each designed CubeSat

		Beamless				Beam			
		C1	C2	C3	C4	C5	C6	C7	C8
Weight		6 th	3 rd	1 st	2 nd	8 th	7 th	5 th	4 th
Space		1 st				2 nd			
Safety Factor	Vertical	8 th	6 th	7 th	5 th	4 th	3 rd	2 nd	1 st
	Horizontal	5 th	6 th	8 th	7 th	2 nd	1 st	3 rd	4 th

The simulation for thermal is concerned with the discharging rate of battery. The discharging rate of the battery will differ depending on the components inside the CubeSat as certain equipments might require more power hence increases the discharging rate of the battery. When the discharge time change, the total power produce by the battery will change and thus effects the internal temperature of the CubeSat. Three studies were made simulating the internal temperature of the CubeSat when the battery discharge time is in (a) 1 hour, (b) 2 hour and (c) 3 hour.

(a) 1 Hour:

The main concerns of conducting analysis in ANSYS Icepak for the CubeSat internal board is to determine the temperature inside the CubeSat when the battery is out of direct sunlight and the battery is in the discharging state. The reason it is important to model the internal temperature for this stage is because, when the CubeSat is out of sunlight, the battery must discharge in order for the internal component to function properly. If the temperature inside is either too low or too hot, the battery may damaged and causes failure. Referring to Figure 14, it can be seen that the highest temperature is located at the battery block, with the value of 38.5954°C. Since the working temperature of battery is -20°C to 60°C, therefore the battery is already in its operating temperature meaning that the battery can function properly when discharging.

Referring to the Table 3, it can be noticed that when the battery discharge time increases the internal temperature will decreases. This situation occur because the battery discharge time will affect the current, I . When the current, I decrease it will causes the power, P to decrease as well because P is dependent on I as the voltage, V is constant. The longer the battery discharge time, the lower the internal temperature of CubeSat. Besides that, as shown in Table 4, heater is needed in the CubeSat unless the battery discharge time is within 1 hour or less.

Table 3- Compile result for thermal subsystem in different battery discharge time

	Battery Discharge Time		
	1 Hour	2 Hour	3 Hour
Temperature (°C)	38.5954	-88.5883	-136.969
Within operating temperature?	Yes	No	No

Conclusion

The three main concerns regarding the mechanical structure of CubeSat are: weight, space and safety factor. Each of the proposed design of the CubeSat has certain advantages and disadvantages hence it is important for the designer to select which type of design is more suitable for their CubeSat mission. For example certain mission might require the reduction of weight for CubeSat hence the design of C4 will be preferred while certain mission might require higher structural strength hence the design of C6 or C7 will be preferred. There is no such thing as best design for the CubeSat structure but only the most suitable design for the mission profile. The main concern of thermal analysis is to determine the internal temperature of the CubeSat when it is in the orbit. The internal temperature of the CubeSat will differ depending on the battery discharge time. The longer the battery discharge time, the lower the internal temperature of CubeSat. Heater is definitely a must inside CubeSat unless the battery discharge time is equal or lower than 1 hour.

Acknowledgment

The authors are gratefully acknowledged the financial support by Malaysia Ministry of Education who supported this project under Research Acculturation Grant Scheme (RAGS) grant no. 9018-00015.

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Mechanical and Materials Engineering

10.4028/www.scientific.net/AMM.554

Stress and Thermal Analysis of CubeSat Structure

10.4028/www.scientific.net/AMM.554.426